LAB ASSIGNMENT – 18.4

Task 1 – Movie Database API Integration

Scenario

You are building a movie recommendation tool that fetches movie details from a public Movie Database API.

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[ ]

```
Generate a Python script using requests to fetch movie details from OMDb API.
Accept movie title as input.
Display title, year, rating, and plot.
Handle errors like invalid movie name, API key issues, and timeout.
```

[2]

```python
import requests

API_KEY = "your_api_key"  # Replace with your OMDb API key

def get_movie(title):
    url = f"http://www.omdbapi.com/?t={title}&apikey={API_KEY}"
    try:
        response = requests.get(url, timeout=5)
        response.raise_for_status()
        data = response.json()

        if data.get("Response") == "False":
            print("Error: Movie not found!")
            return

        print("\nMovie Details")
        print("Title:", data.get("Title"))
        print("Year:", data.get("Year"))
        print("Rating:", data.get("imdbRating"))
        print("Plot:", data.get("Plot"))

    except requests.exceptions.Timeout:
        print("Error: Request timed out!")
    except requests.exceptions.RequestException as e:
        print("Error:", e)

title = input("Enter movie name: ")
get_movie(title)
```

```
Enter movie name: ishq
```

Task 2 – Public Transport Schedule API

Scenario

A city transport app needs to fetch bus/train arrival timings from a transport API.

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```
Generate Python code to fetch public transport arrival times using a REST API.
Accept stop ID as input.
Display next 3 arrivals.
Handle invalid input, server errors, and timeout.
Implement retry logic with at least 1 retry.
```

```python
import requests
import time

def get_arrivals(stop_id):
    url = f"https://api.example.com/arrivals/{stop_id}"

    for attempt in range(2):  # Retry once
        try:
            response = requests.get(url, timeout=5)
            response.raise_for_status()
            data = response.json()

            arrivals = data.get("arrivals", [])
            if not arrivals:
                print("No arrival data available")
                return

            print("\nNext 3 Arrivals:")
            for i in arrivals[:3]:
                print(i)

            return

        except requests.exceptions.Timeout:
            print("Timeout! Retrying...")
        except requests.exceptions.RequestException:
            print("Server error!")
            return

        time.sleep(2)

stop_id = input("Enter stop ID: ")
if stop_id.strip() == "":
    print("Invalid Stop ID")
else:
    get_arrivals(stop_id)
```

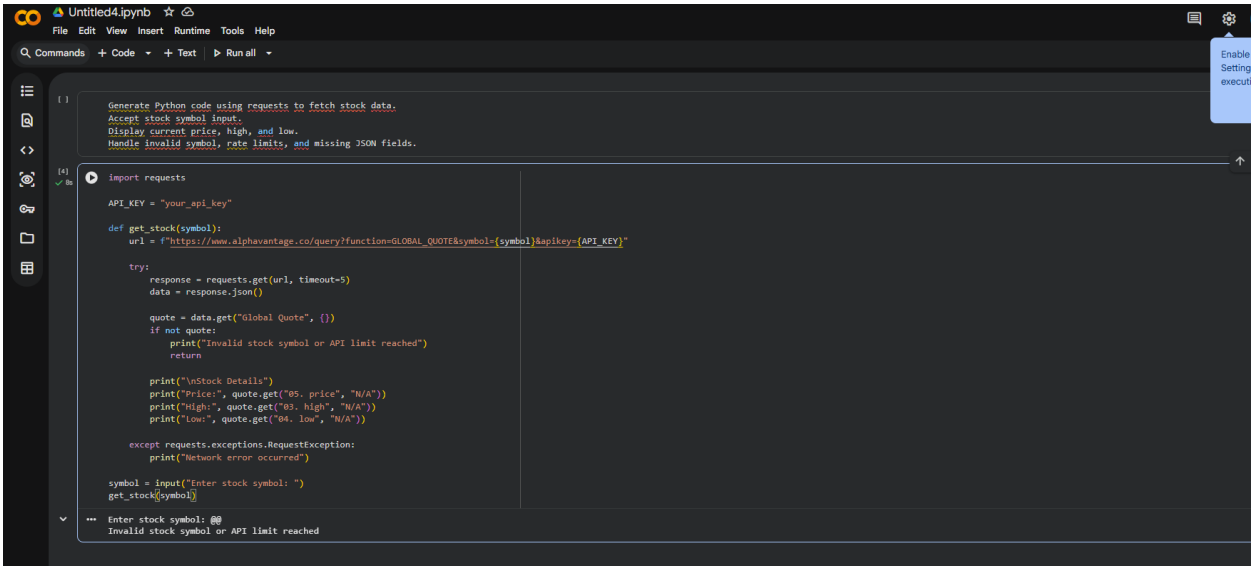
```
Enter stop ID: 1234
Server error!
```

Task 3 – Stock Market Data API

Scenario: You are developing a finance dashboard that retrieves live stock prices.

Task

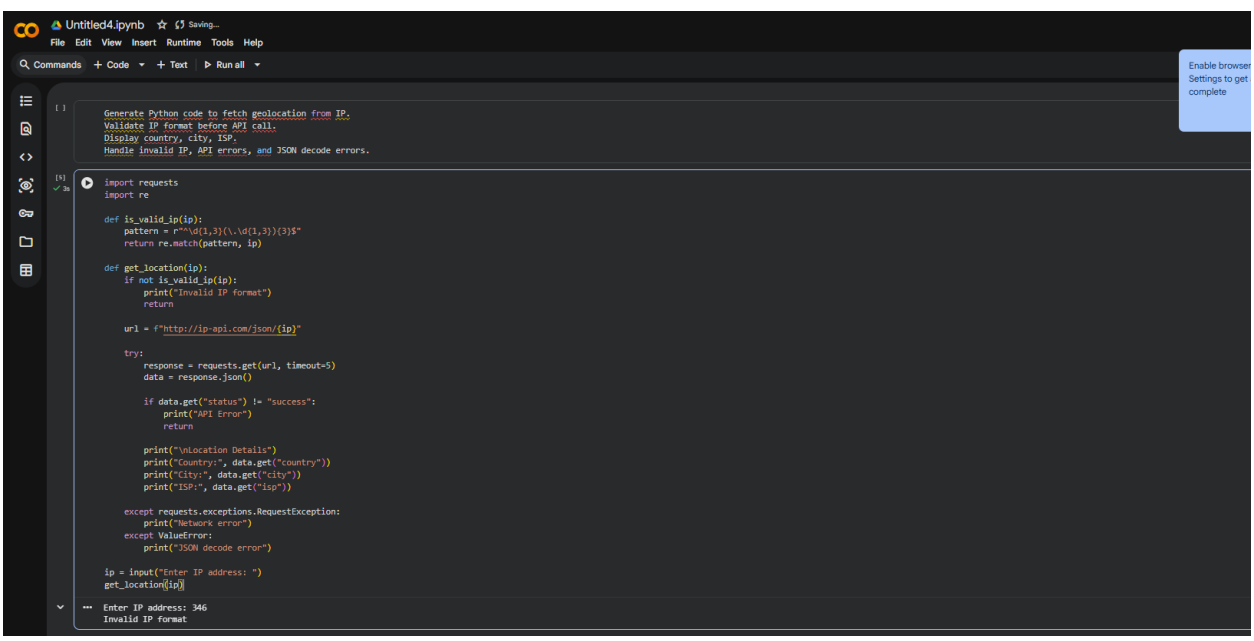
Use AI to generate a Python script that fetches stock data from a financial API.



Task 4 – Geolocation API (IP AddressLookup)

Scenario

A cybersecurity tool needs to detect the geographical location of a user based on IP address.



Task 5 – Real-Time Application: Payment Gateway Simulation API

Scenario

You are building a test payment integration system that connects to a sandbox payment API.

Untitled4.ipynb

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```
Generate Python code to simulate payment using POST request.
Accept amount, currency, payment method.
Validate inputs.
Handle HTTP errors (400, 401, 500), network issues.
Log failed transactions to a file.
```

```python
import requests

def process_payment(amount, currency, method):
    url = "https://api.example.com/payment"

    if amount <= 0:
        print("Invalid amount")
        return

    payload = {
        "amount": amount,
        "currency": currency,
        "method": method
    }

    try:
        response = requests.post(url, json=payload, timeout=5)

        if response.status_code == 200:
            print("Payment Successful")
        elif response.status_code in [400, 401, 500]:
            print("Payment Failed:", response.status_code)
            with open("failed_log.txt", "a") as f:
                f.write(str(payload) + "\n")
        else:
            print("Unexpected error")

    except requests.exceptions.RequestException:
        print("Network error occurred")
        with open("failed_log.txt", "a") as f:
            f.write(str(payload) + "\n")

amount = float(input("Enter amount: "))
currency = input("Enter currency: ")
method = input("Enter payment method: ")

process_payment(amount, currency, method)
```

```
Enter amount: 34000
Enter currency: dollar
Enter payment method: phonepay
Network error occurred
```